

KHALID AHMAD

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Research Interests

Applied Machine Learning, Computer Vision, Natural Language Processing, AI4Health

Education

Federal University, Lokoja

Sep. 2018 – Oct. 2023

B.Sc. Computer Science CGPA: 4.75/5.0 (3.9/4.0)

Lokoja, Nigeria

Research Experience

Federal University, Lokoja

Oct 2022 – Present

Research Assistant

Lokoja, Nigeria — Hybrid

- Conducting a literature review for the development of an AI-based system for early detection of plant diseases using Vision Transformers and multispectral imagery.

Publications

- **Khalid, H.A.**, Abdulhakeem A.O., & Aifuobhokhan, J. (Under Review). PERG-EyeNet: An Interpretable Boosting Framework for Automated Classification of Pattern Electroretinogram Signals with Clinical Metadata. *Journal of Healthcare Informatics Research*.
- Aifuobhokhan, J., **Khalid, H.A.**, Ekechi, C.C., Olanrewaju, A.O., Afuadajo, E., Bowale, D.A., & Inioluwa, O.M. (Under Review). Predicting Thyroid Cancer Recurrence Using Machine Learning: An Artificial Intelligence Approach to Clinical Oncology. *International Journal of Advances in Artificial Intelligence and Machine Learning*.
- Aderemi, I.A., Kehinde, T.O., Ugochukwu, D.O., **Khalid, H.A.**, Adjei, K.Y., & Chijioke, C.E. (2025). Beyond the Black Box: A Systematic Review of Explainable AI for Transparent and Trustworthy Water Quality Monitoring. *IEEE Sensors Reviews*, August 2025. [DOI]
- **Khalid, H.A.**, & Fati, O.O. (2025). A real-time road anomaly detection via masked autoencoder-enhanced vision transformers. *International Journal of Innovative Research in Engineering & Management*. [DOI]
- Kossy, S., Saidu, M., **Khalid, H.A.**, Ogeah, P.U., Bakare-Abidola, T., Aanuoluwa, T.I., & Obafemi, A.B. (2025). Advancing environmental sustainability through emerging AI-based monitoring and mitigation strategies for microplastic pollution in aquatic ecosystems. *World Journal of Biology Pharmacy and Health Sciences*. [DOI]

Conferences & Presentations

- **Khalid, H.A.**, Fati, O.O., Faruk, A., & Ismail, A. Random forest-based predictive model for estimating first-year GPA in Nigerian tertiary institutions.
- **Khalid, H.A.**. Exploring Technology Hotspots: The Impact of Mathematics on Computer Science. *National Association of Computing Students (NACOS) Workshop, FUL Chapter*, Aug 2022
- **Khalid, H.A.**, Musa, A., & Grace, A. Four Ways Traffic Light Design and Implementation. *Department of Computer Science Symposium, Federal University Lokoja*, Oct 2021

Academic & Professional Service

Deep Learning Indaba

Jul 2025 – Aug 2025

Judge

Kigali, Rwanda — Hybrid

- Selected as a judge for the Deep Learning Indaba 2025 AI Community Challenge in Kigali to support the evaluation of innovative AI projects from over 160 teams across more than 20 countries.
- Supported the advancement of real-world AI impact by promoting emerging research and recognizing talent within the African AI community.

Technical Skills

Languages & Frameworks: Python, C++, JavaScript, HTML, CSS, SQL, Flask, Bootstrap, Streamlit

Machine Learning & AI: PyTorch/TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, NLTK, Matplotlib

Developer Tools & Other Technologies: Git, Docker, VS Code, Jupyter, GitHub, Slack, Azure, MySQL, Hugging Face

Writing Tools: LaTeX, Microsoft Word, Google Docs.

Teaching Experience

Digital onboarders

Jan 2024 – Nov 2024

CS Mentor & AI Workshop Facilitator

Lafia, Nigeria — Hybrid

- * Delivered hands-on instruction in Python and core CS subjects to 75+ students, guiding them in developing web applications and mini-projects.
- * Led 25+ lab sessions using Jupyter Notebooks and GitHub; graded 50+ student projects with personalized feedback to foster learning.
- * Participated in 3+ national workshops and the Miva Conference on AI, contributing to discussions on AI and Nigeria's digital economy.

Federal University, Lokoja

Mar 2019 – Jul 2023

Academic Tutor

Lokoja, Nigeria — Onsite

- * Tutored over 2,500 students in core STEM subjects including Physics, Mathematics, Statistics, and Programming, covering topics like Linear Algebra, Calculus, and Probability.
- * Led 30+ exam prep sessions and 80+ personalized 1-on-1 tutorials in Python and C++, contributing to a 20–35% documented improvement in student performance.
- * Assisted 5+ faculty members in teaching CS courses by simplifying complex topics in Operating Systems, Compiler Design, Discrete Structures, and Machine Learning, contributing to a 15% average improvement in student assessment scores.

Cali Vision Academy

Feb 2021 – Sept 2022

ICT Instructor

Lokoja, Nigeria — Onsite

- * Taught over 120 secondary school students IT fundamentals, productivity tools, and introductory programming, resulting in 85% of students earning a distinction (A) in WAEC Computer Studies.
- * Developed and delivered a structured 3-term ICT curriculum with 24 lesson plans and 50+ practical exercises, improving average student performance by 35%.
- * Organized 2 intra-school ICT competitions that increased STEM club participation by 60% and enhanced student engagement in technology.

Industrial Experience

Omdena

Feb 2025 – May 2025

Machine Learning Engineer

New York, USA — Remote

- * Collaborated with a global team of 10+ engineers on a Smart Farming project to boost yields for 10,000+ farmers in Kano State, Nigeria.
- * Cleaned and engineered features from 148,666 soil moisture records to enhance model inputs, improving data quality and system reliability.
- * Trained a Random Forest model that achieved 99% precision in classifying plant health indicators, improving detection accuracy by 9% over baseline.

Saint Louis University

Jun 2023 – Jul 2023

Data Analyst

Missouri, USA — Remote

- * Analyzed digital marketing campaign data for Finance and Marketing teams, tracking CPC, CPR, reach, and engagement metrics across 10 campaigns.
- * Built interactive dashboards using Python, SQL, and Power BI to visualize performance insights, identifying top-performing campaigns (2 and 8).
- * Reported ad spend of INR 12,088.61, reaching 188,868 users and generating 5,257 unique link clicks; recommended optimizations for underperformers (3 and 10).

Nigerian Midstream & Downstream Petroleum Regulatory Authority

Mar 2022 – Aug 2022

Data Scientist Intern

Lokoja, Nigeria — Onsite

- * Analyzed 150+ petrol samples from 40+ Lokoja filling stations, uncovering a 22% adulteration rate and informing regulatory action.
- * Built ML models (Logistic Regression, Decision Trees, Random Forest, SVM) with 75% accuracy in detecting fuel adulteration using preprocessed data in Python.
- * Reduced manual inspection workload by 30% by integrating predictive analytics into existing regulatory workflows.

Al-Harir Global Services

Jan 2021 – Jan 2022

Frontend Developer

Lokoja, Nigeria — Hybrid

- * Collaborated with UI/UX designers to build a responsive and visually appealing marketing landing page, improving user engagement by 20%.
- * Implemented cross-browser responsive layouts using HTML, CSS, and JavaScript, enhancing site accessibility across mobile and desktop platforms.

Leadership & Service

Senator Representative

Federal University Lokoja

National Association of Lokoja Local Government Students (NALLGS)

May 2020 – Jun 2022

- * Represented interests of over 50 students from Lokoja Local Government at national academic forums, influencing key policy discussions.
- * Facilitated communication between 50+ students and association leadership, ensuring timely resolution of student concerns.

General Secretary

NACOS Electoral Committee

Nigeria Association Of Computing Students.

June 2022

- * Coordinated the organization and execution of elections involving 5,000+ student voters, increasing participation by 15%.
- * Ensured transparency and fairness by maintaining accurate records and overseeing electoral procedures, reducing disputes by 30%.

Honors & Awards

- * Wale Lab Research Internship – 2025: Fully funded AI healthcare research internship.
- * Departmental and Faculty Fee Waiver – 2023: Awarded for top academic performance and engagement.
- * Ayomide Ajiboye Scholar (USA) – 2023: Scholarship and mentorship for academic excellence.
- * Student Tutor of the Year – 2023: Recognized for outstanding peer tutoring.
- * Department Contribution Award – 2022: Honored for impactful research and academic service.
- * NMDPRA Recommendation – 2022: Commended for exceptional work with Nigeria's petroleum authority.
- * Dean's Honor Award – 2020-2023: Multiple-year award for sustained academic excellence.
- * Kogi State Bursary Scholar – 2020: Financial award for academic merit and need.

Volunteering Experience

Digital Onboarders (DODEEL)

Feb 2024 – Aug 2024

Community Development Service (CDS) Volunteer

Lafia, Nigeria — Onsite

- Led digital literacy sensitization programs in 5+ Government Secondary Schools in Lafia, teaching foundational computer skills to over 300 students.
- Mentored 50+ students in Lafia on personal development and digital skills awareness.
- Trained 40+ small business owners in Lafia on using Google My Business and Maps to improve their digital visibility.

National Youth Service Corps (NYSC)

Nov 2023 – Dec 2023

Data Entry Analyst

Edo, Nigeria — Onsite

- Processed and verified records for over 3,500 NYSC Batch C2 corps members deployed across Edo State.
- Conducted exploratory data analysis (EDA) to extract demographic insights on gender distribution, educational background, and skill acquisition program participation, aiding program planning and reporting.

United Nations International Children's Emergency Fund (UNICEF)

Aug 2023 – Dec 2023

Data Entry Analyst

Lokoja, Nigeria — Hybrid

- Collected data from over 500 households, 3 hospitals, and 2 government agencies in Lokoja to support accurate population health reporting.
- Registered more than 1,200 newborns into UNICEF's database and facilitated the distribution of 1,200+ birth certificates to local healthcare facilities.

Federal University Lokoja

Mar 2021 – Oct 2023

Social Media Moderator

Lokoja, Nigeria — Hybrid

- Co-founded and managed a student-led social media support group with 80,000+ members at Federal University Lokoja, dedicated to assisting new and returning students with academic, housing, and campus-related inquiries.
- Moderated discussions and responded to hundreds of student inquiries weekly, fostering a trusted and inclusive digital community.
- Developed and shared informational content that improved student onboarding, reduced confusion during registration periods, and boosted engagement across the platform.

Professional Affiliations

- Member, Black in AI (BAI)
- Member, Data Science Nigeria (DSN)
- Member, National Association of Computing Students (NACOS)
- Member, Nigeria Computer Society (NCS)
- Core Member, Google Developer Student Clubs (GDSC)

Referees

- Available upon request